

ASSIGNMENT - X

X. Application of Laplace Transforms to Differential Equations, Simultaneous Differential Equations :

Solve the following using Laplace Transforms :

1. $y''' - 3y'' + 3y' - y = t^2 e^t; y(0) = 1; y'(0) = 0; y''(0) = -2$

Ans : $y = e^t - te^t - \frac{t^2 e^t}{2} + \frac{t^5 e^t}{60}$

2. $ty'' + (1 - 2t)y' - 2y = 0; y(0) = 1; y'(0) = 2$

Ans : $y = e^{2t}$

3. $ty'' + 2y' + ty = \sin t; y(0) = 1$

Ans : $y = \frac{1}{2} \left[\frac{3 \sin t}{t} - \cos t \right]$

4. $\frac{dx}{dt} - y = e^t; \frac{dy}{dt} + x = \sin t; x(0) = 1; y(0) = 0$

Ans : $x = \frac{1}{2} (e^t + \cos t + 2 \sin t - t \cos t)$

$y = \frac{1}{2} (t \sin t - e^t + \cos t - \sin t)$

5. $\frac{dx}{dt} - 6x + 3y = 8e^t; \frac{dy}{dt} - 2x - y = 4e^t; x(0) = -1; y(0) = 0$

Ans : $x(t) = -2e^t + e^{4t}; y(t) = -\frac{2}{3}e^t + \frac{2}{3}e^{4t}$